Supplementary Material for: Successfully Defending Academic Integrity: an Ensemble Approach Towards Detecting Machine-generated Texts

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A Dataset Information

?? gives an overview about the different datasets we incorporate. For each dataset, their amount of human and generative artificial intelligence sentences is given. Also the licenses under which they were made available is shown.

The self-generated datasets were created as follows:

AI papers: We searched for the last 100 papers within these categories on arXiv:
- Astrophysics
- General finance
- Biology

[1] Number of human sentences
[2] Number of generative AI fabricated sentences
[3] https://doi.org/10.1109/TAI.2025.3544183 and https://github.com/rsickle1/human-v-ai
[4] https://www.kaggle.com/datasets/denvermagtibay/ai-generated-essays-dataset
[5] https://www.kaggle.com/datasets/prince7489/ai-vs-human-comparison-dataset
[6] https://www.kaggle.com/datasets/pratyushpuri/ai-vs-human-content-detection-1000-record-in-2025
[7] https://www.kaggle.com/datasets/shanegerami/ai-vs-human-text/data
[8] https://www.kaggle.com/datasets/gulfan/ai-vs-human-text-classification-dataset
[9] https://www.kaggle.com/datasets/zunorain/balanced-ai-human-humanized-updated
[10] https://www.kaggle.com/datasets/navjotkaushal/human-vs-ai-generated-essays
[11] https://github.com/shamylafirdoos/Gpt-vs-Human-Text-Classification
[12] https://arxiv.org/abs/2510.22874 and https://huggingface.co/datasets/Rajarshi-Roy-research/Defactify_Text_Dataset

Table 1: Data sources used for our dataset.

Name	HS[1]	GAIS[2]	Comment	Reference	License
5000 human 5000 machine	64,170	68,876	n/a	GitHub[3]	CC BY 4.0
AI generated Text Dataset	38,032	799	n/a	Kaggle[4]	Public domain
AI vs Human Comparison Dataset	360	347	n/a	Kaggle[5]	Public domain
AI vs Human Content Detection 1000+ record in 2025	17,128	17,828	n/a	Kaggle[6]	Apache 2.0
AI vs. Human text	154,846	0	Only 2.375% of human sentences are included	Kaggle[7]	Other (combined from previous public datasets)
AI vs. Human Text Classification Dataset	727	648	n/a	Kaggle[8]	Public domain
AI papers	0	79,023	n/a	self-generated	OpenAI terms of use
AJUR	48,919	0	n/a	self-generated	Open source
Balanced_ AI_Human_ Humanized_UPDATED	6,616	5,065	Humanized data is omitted	Kaggle[9]	Apache 2.0
Human vs AI Generated Essays	38,032	2,089	n/a	Kaggle[10]	Public domain
Intersect	10,318	0	n/a	self-generated	Open access
Shuffled	67,828	39,680	n/a	GitHub[11]	MIT
Train	235,549	461,040	Human, Gemma, Llama and GPT only	HuggingFace[12]	CC BY 4.0
Wikipedia	3,312	5,503	n/a	self-generated	GFDL/CC BY-SA 4.0
Sum	**685,837**	**680,898**	n/a	n/a	n/a
Train data	480,086	476,628	Used to train the models	n/a	(see above)
Validation data	102,876	102,135	Used to validate the models	n/a	(see above)
Test data	102,875	102,135	Used to test the models	n/a	(see above)

The goal was finding papers in topics which are away from the course we want to use the model for (Internet of Things area) to prevent potential biases. We only collected the titles of the paper. Once we received the paper titles, we sent a query to OpenAI's Generative Pretrained Transformer 5.1 asking it to generate a paper matching the title. The dataset only consists of Generative Pretrained Transformer's responses.
Each query was sent using OpenAI's batch Application Programming Interface[13] with these parameters:

System query: 'Create a scientific paper for the given title. It should have a length of 12 DIN a4 pages. Don't ask the user anything. Always write full sentences and include a literature review.'

User query The title of the paper searched from ArXiV.

AJUR: We downloaded all papers submitted to the Americal Journal of Undergraduate Research[14] published before 2022. Afterwards we mined the text of these papers and removed irrelevant data (references, author information, ...). Our assumption is that adding student texts to our dataset helps the model in correctly classifying text authored by students as human written.

Intersect: We used the same procedure as for AJUR but for the Intersect Journal[15].

Wikipedia: We selected the Wikipedia articles for the computer science category from the dump from 2021-01-01. People, companies and software names were omitted in order to focus only on terminology. For each article we considered only the first paragraph as human data. Furthermore, we collected the titles of the articles. For each title, we queried OpenAI's Generative Pretrained Transformer 5.1 using the batch Application Programming Interface to generate a short description for the provided term. The idea behind this is that the student submissions we receive contain definitions. Having a dataset with a human definition and its generative artificial intelligence counterpart should help to figure out the author.
We used these parameters to query OpenAI:

System query: 'Give a short description of the provided term (1-2 paragraphs). Don't ask the user anything.'

User query: The title of the Wikipedia article.

B Model parameters

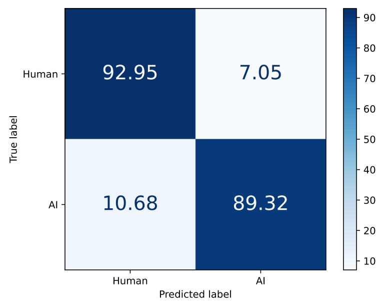
- Support vector machine:
  Scaler: StandardScaler
  Kernel: RBF
  C: 1.0
- FFNN:
  Loss function: Cross-entropy

[13] `https://platform.openai.com/docs/guides/batch`
[14] `https://ajuronline.org/`
[15] `https://ojs.stanford.edu/ojs/index.php/intersect/index`

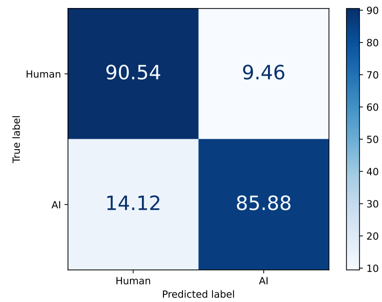
  Dropout: 0.5
  Optimizer: **Type:** AdamW
    Learning rate: 2×10^{-4}
    Weight decay: 5×10^{-4}
  Scheduler: **Type:** Reduce learning rate on plateau
    Factor: 0.5
    Patience: 2
- RoBERTa:
  Max length: 256
  Optimizer: **Type:** AdamW
    Learning rate: 3×10^{-5}
- DistilBERT:
  Max length: 256
  Optimizer: **Type:** AdamW
    Learning rate: 3×10^{-4}

The mentioned parameters were found through manual adjustments and evaluations.

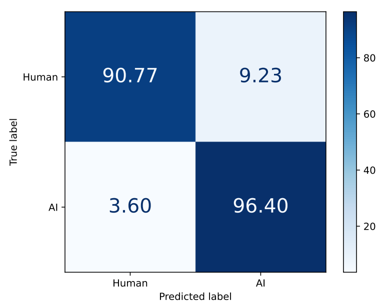
C Confusion Matrices of the Individual Classifiers



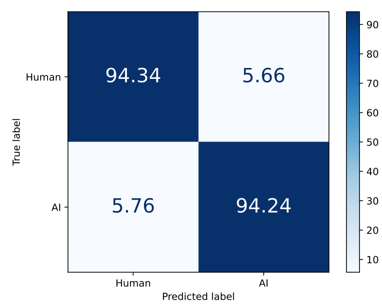
(a) SVM confusion matrix.



(b) FFNN confusion matrix.



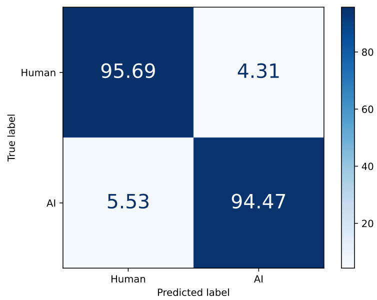
(c) DistilBERT confusion matrix.



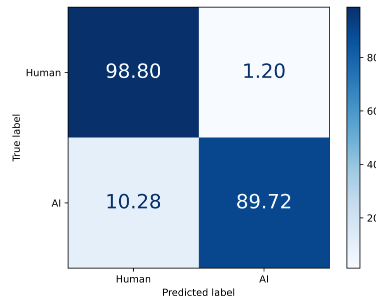
(d) RoBERTa confusion matrix.

Fig. 1: Confusion matrices of individual classifiers.

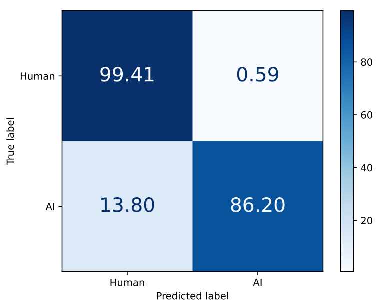
D Boosting Confusion Matrices



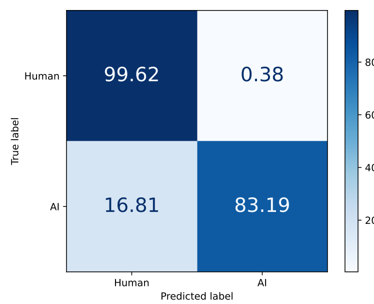
(a) 50% cutoff



(b) 75% cutoff



(c) 90% cutoff



(d) 95% cutoff

Fig. 2: Confusion matrices for different cutoffs.